

● Off-road autonomy chassis

LUCHS B

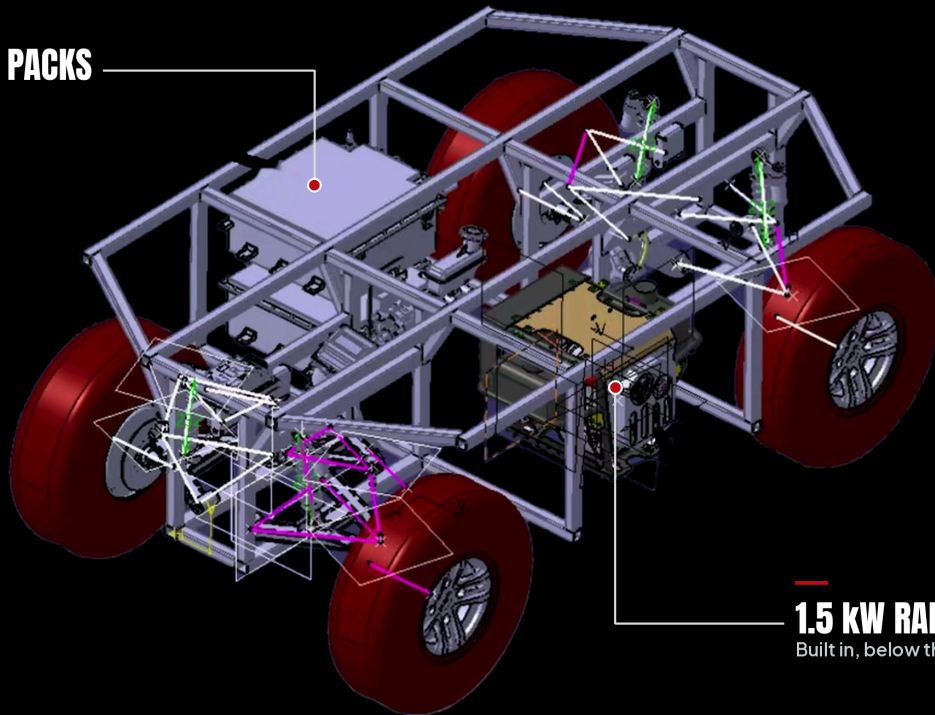
USER MANUAL

Off-road autonomy chassis · Standard and Pro

Preliminary edition · 2026-10-05

TWO BATTERY PACKS

About 5 kWh each



1.5 kW RANGE EXTENDER

Built in, below the deck



VERSION 0.9.5 · PRELIMINARY

October 2026 · English edition

CHASSIS BY RJ TECH LIMITED

Autonomy by TerraAgent Robotics

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4. **Force majeure.** Fire, earthquake, flood, lightning, war, terrorism, cyber-attack, government action or any other event beyond the manufacturer's reasonable control.
5. **Normal wear and ageing.** Loss of performance caused by normal wear and ageing under normal operating conditions.

The Luchs B is a chassis for building autonomous and remotely operated vehicles. It is not approved for use on public roads. You are responsible for the safety case of the complete vehicle you build on it, including your software and any payload.

Revision history

Version	Date	Changes
0.9.5	2026-10-05	Preliminary edition.

About this manual

Thank you for choosing the Luchs B, a full-size off-road chassis for autonomous and remotely operated vehicles.

This manual explains how to operate the chassis safely and how to control it from your own computer. Please read it before you start, and keep it for future reference.

Who this manual is for

- **Autonomy and robotics engineers** who integrate sensors, compute and software on the Luchs B and need the interface, the control modes and the safety behaviour in detail.
- **Operators** who drive the vehicle by remote control, charge it and carry out daily checks.
- **Service engineers** who commission, maintain and troubleshoot the chassis.

Software versions

This manual describes chassis firmware that implements the Luchs Chassis Interface (LCI) v1.0. This firmware is in development. Units shipped before it is released use the VCU interface V2.6, described in section 5.3. The ROS software (section 5.4) supports both.

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SAFETY



READ BEFORE USE

The Luchs B weighs about 230 kg and can reach 40 km/h. A vehicle of this size can cause serious injury. Read and understand all safety information in this manual before operating it.

Prohibited

Environment

- Do not operate the chassis on public roads.
- Do not charge the chassis in rain or snow, or anywhere the charging inlet may come into contact with water.
- Do not operate the chassis in the presence of flammable or explosive gases.
- Do not operate or store the chassis outside its operating temperature range of -20°C to $+55^{\circ}\text{C}$.
- Do not drive through water deeper than 300 mm.

Operation

- Do not allow children, untrained persons, or anyone who is tired or under the influence of alcohol or medication to operate the chassis.
- Do not ride on the chassis or let anyone ride on it.
- Do not stand in front of or behind the chassis while it is powered, and keep a clear zone of at least 5 m around it while it moves.
- Do not exceed the maximum payload of 200 kg, or the working payload of 80 kg for continuous operation.
- Do not disassemble or service the chassis, or touch moving or hot parts (wheels, motors, brakes, power electronics), while it is switched on.

Charging

- Use only the charging equipment approved for the chassis. An unsuitable power supply can cause fire.
- Charge only at ambient temperatures between 0°C and 55°C .
- If the battery overheats, smokes or smells unusual, stop charging immediately and move away.

Mandatory

Before each use

- Walk once around the vehicle. Check the tyres, the bumpers, the deck load and its fixings, and that nothing is caught in the wheels.
- Check that the emergency stop button and both bumpers stop the vehicle (section 4.5), and that the transmitter battery is charged.
- When testing your own software, keep the transmitter in hand and a safety operator ready to stop the vehicle.

Charging and maintenance

- Charge in a dry, well-ventilated place, away from flammable materials.
- Use only original spare parts. If the chassis will not be used for a long time, keep the battery at 50–70 % and charge it regularly.
- If any fault occurs, switch the chassis off and contact support. Do not attempt to repair high-voltage parts yourself.

In an emergency

- 1 Press the emergency stop on the side panel, or flip switch A on the transmitter.
- 2 Switch the chassis off.
- 3 Follow your emergency plan, move people to safety and deal with the situation.



PRODUCT OVERVIEW

2.1 Introduction

The Luchs B is a 4×4 off-road chassis designed to carry an autonomy stack. Four in-wheel motors, two electric power-steering units, an electro-hydraulic brake and an electric parking brake are all controlled by the chassis controller. Your computer sends motion commands over one CAN interface; the chassis executes them, protects itself and reports its state.

The Luchs B speaks the Luchs Chassis Interface (LCI). Its core frames use the same identifiers and layouts as the protocol of the smaller Luchs A, and one ROS software package drives both chassis: software written for one runs on the other with a change of vehicle parameters.

2.2 Key features

- **Off-road capable.** 300 mm class ground clearance, independent suspension, 22-inch tyres, 4WD with 330 N·m peak torque per wheel.
- **Steering on both axes.** Front Ackermann for stable high-speed driving, dual Ackermann for tight turns, selected by command.
- **Built to be driven by a computer.** Motion-level commands, odometry-grade feedback, live limits, a host heartbeat and a documented DBC.
- **Layered safety.** Emergency stops on the vehicle, the bumpers, the transmitter and the host interface; a parking brake that holds a full load on a 20° slope.

2.3 Specifications

Table 2-2 · Technical data

SIZE AND WEIGHT	
Dimensions (L×W×H)	1990 × 1150 × 800 mm
Wheelbase	1430 mm
Track width	970 mm
Ground clearance	300 mm
Tyres	22×7-10 in off-road
Chassis weight	230 kg
Working / maximum payload	80 kg / 200 kg
Suspension travel	50 mm, independent at each wheel
Versions	Black: standard · White: cooling version with heat-protective coating · Range extender built in (exposed on the deck only on test vehicles)
Approach / departure / ramp angle	90° / 90° / 40°
Maximum side tilt (full load)	30°
MOTION AND PERFORMANCE	
Drive	4WD, 2WD selectable · 4 in-wheel motors
Steering	Dual EPS · front Ackermann or dual Ackermann, selectable

Maximum road-wheel angle	30°
Steering resolution (wheel end)	0.1°
Minimum turning diameter	4.4 m (dual Ackermann)
Top speed (full load)	40 km/h
Maximum acceleration (no load)	0.8 g
Maximum climb (full load)	30° (58 %)
Maximum side slope while driving	40 % (22°)
Vertical step / trench / wading	150 mm / 400 mm / 300 mm
Service brake	Electro-hydraulic, 240 ms response
Parking brake	Electric parking brake; holds full load on 20° for ≥5 min
Creep mode	Yes
DRIVE AND POWER	
Drive motors	4 × in-wheel, 72 V · 1.8 kW rated / 5 kW peak each
Torque per wheel	130 N·m rated / 330 N·m peak
Peak system power	20 kW
Battery	LiFePO ₄ · 72 V 60 Ah (4.3 kWh) or 73 V 127.4 Ah (9.3 kWh) · >1000 cycles
Range	≥60 km (full load, 20 km/h, hard dirt) · ≥100 km (no load, 40 km/h, paved) · up to 300 km with range extender
Desert mode (option)	Two battery packs, about 5 kWh each, plus a 1.5 kW range extender built into the chassis · up to 300 km ·
Charging	On-board charger, full in 3 h · battery swap in under 3 min · AC inlet type per market
Payload power	12 V / 24 V switched outputs, fused
CONTROL AND COMMUNICATION	
Operating modes	Remote control · Automatic · Maintenance (planned)
Interface	LCI v1.0 over a dedicated CAN 2.0B channel, 1 Mbit/s, big-endian
Software	ROS 2 Humble / Jazzy and ROS 1 Noetic (luchs_ros 1.1: SDK, driver, robot model, simulator; one code base with Luchs A) · LCI and VCU protocol documentation · DBC files
Units before LCI firmware	VCU interface V2.6 over CAN, supported by the same software
Chassis IMU	Six-axis, 100 Hz
Remote control	2.4 GHz handheld transmitter, included
Lighting	Headlights, turn indicators, brake and reverse lights · stealth mode switches all lights off
SAFETY	
Emergency stop	Side-panel button · front and rear bumper strips · remote control · host command · external hard-wired loop (planned)
Protection	Power-on self-test and diagnostics · battery management · host heartbeat · live limits
ENVIRONMENT	

Operating temperature	-20 °C to +55 °C
Humidity	95 % (±3 %), tested at 20–30 °C
Ingress protection	IP67 below 440 mm (battery, motors, connectors) · IP65 for controllers
Maximum altitude	5000 m
Battery transport	UN38.3

Typical values. Speed, range and gradeability depend on payload, surface, temperature and driving style.

2.4 Chassis components



Figure 2-1 · Luchs B black standard version (left) and white cooling version (right). The box on the deck of the white unit is the range extender in its test position; on delivered units it sits inside the chassis.

Side control panel

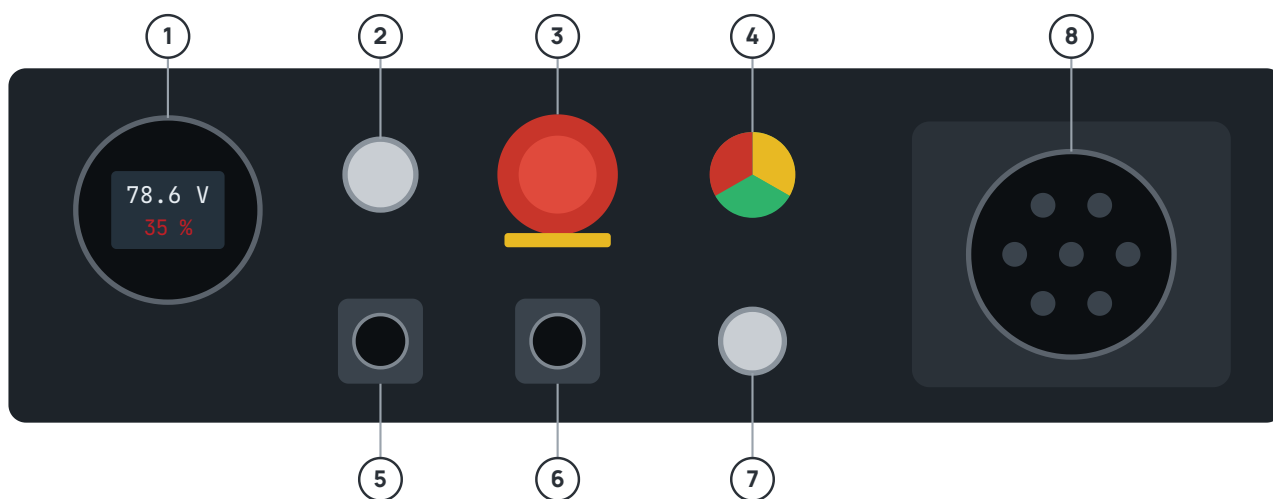


Figure 2-2 · Side control panel

- | | | |
|--|----------------------|---------------------------|
| 1 Battery display (voltage, charge, current) | 3 Emergency stop | 6 24 V payload power port |
| 2 Start button | 4 Charge indicator | 7 Payload power switch |
| | 5 Service / CAN port | 8 AC charging inlet |



Figure 2-3 · White unit with its side panel. Range extender on the deck in test position.

Desert mode (range extender)

Desert mode is the ultimate-endurance configuration: two battery packs of about 5 kWh each and a 1.5 kW range extender, all inside the frame, below the deck. The range extender charges the batteries while you drive and extends the range to up to 300 km. In photos and films of test vehicles you may see the range extender bolted to the deck; that is a test position for easy access.

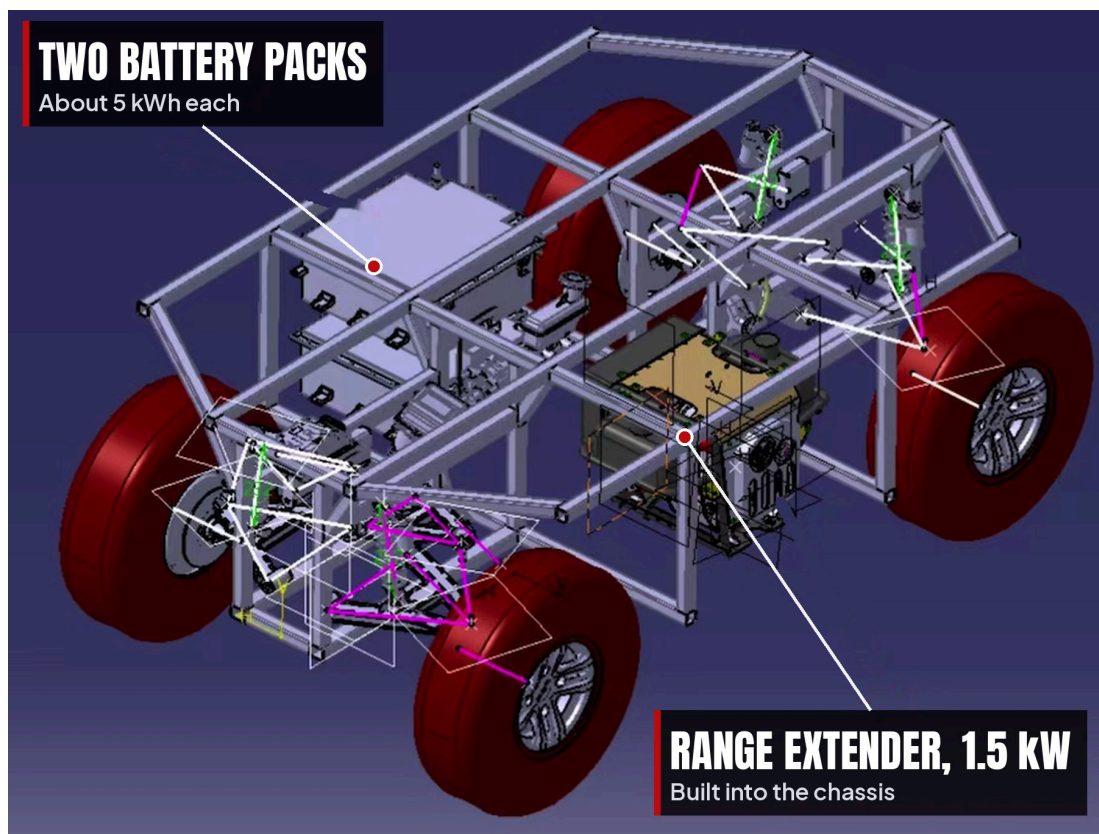


Figure 2-4 · Desert mode in CAD: two battery packs and the 1.5 kW range extender inside the frame.

The charging inlet shown is the type fitted to current units (250 V, 32 A); export units may use a different inlet.

2.5 Remote-control transmitter

The chassis is supplied with a 2.4 GHz handheld transmitter and a matching receiver fitted to the vehicle.

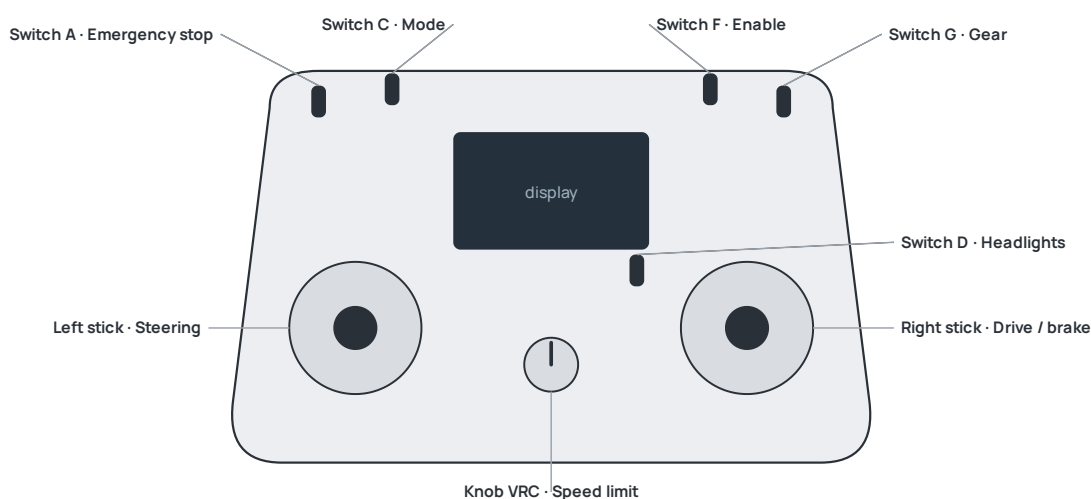


Figure 2-5 · Transmitter controls used with the Luchs B

Table 2-1 · Transmitter controls

Control	Function	See section
Switch F	Enable. Down: transmitter enabled. The vehicle ignores the transmitter while it is not enabled.	4.3
Switch C	Control source. Up: remote control. Down: automatic (host computer).	4.4

Control	Function	See section
Switch G	Gear. Up: neutral. Centre: forward. Down: reverse.	4.3
Switch A	Emergency stop. Down: emergency stop.	4.5
Switch D	Headlights. Down: on.	4.6
Left stick, left / right	Steering. Full deflection = maximum road-wheel angle, 42° (current firmware).	4.3
Right stick, up / down	Up: drive. Down: brake, proportional up to full brake pressure (6 MPa).	4.3
Knob VRC	Speed limit, from 0 to 30 km/h (current firmware; 40 km/h planned).	4.3

NOTE

Do not change the transmitter's channel settings. The vehicle is configured for the settings it was shipped with; changed settings can cause loss of control.



QUICK START

Follow these steps to start your Luchs B safely. Before you begin, make sure you have read and understood Chapter 1.

3.1 Daily check

- 1 **Walk around the vehicle.** Check the area is clear, the tyres are inflated and undamaged, and any load on the deck is secured.
- 2 **Check the emergency stop.** Make sure the emergency stop button is released (turn it clockwise to release).
- 3 **Check the transmitter.** Make sure it has battery and that switch A (emergency stop) is up and switch G (gear) is in neutral.

3.2 Power on and drive by remote control

- 1 **Switch on the chassis.** Press the start button on the side panel. The battery display shows voltage, state of charge and current.
- 2 **Switch on the transmitter.** Push its power switch up; the screen lights up.
- 3 **Select remote control.** Set switch C up.
- 4 **Enable.** Set switch F down.
- 5 **Set a speed limit.** Turn knob VRC up from zero. Start low.
- 6 **Select a gear.** Switch G to centre for forward, or down for reverse.
- 7 **Drive.** Push the right stick up to drive and pull it down to brake. Steer with the left stick.

3.3 Power off

- 1 **Stop the vehicle** and set switch G to neutral. The parking brake applies automatically when no control source is active.
- 2 **Switch off the chassis** with the start button. The battery display goes dark.
- 3 **Switch off the transmitter.**

3.4 Charging

- 1 **Switch off the chassis.** The vehicle must be off while charging.
- 2 **Connect.** Open the cover of the charging inlet on the side panel and plug in the charging connector firmly.
- 3 **Charge.** The charge indicator shows progress. A full charge takes about 3 hours.
- 4 **Finish.** Unplug the connector and close the inlet cover.

3.5 Connecting your computer

- 1 **Mount and wire your equipment** on the deck. Connect your computer's CAN interface to the customer CAN port, and power your equipment from the payload power port (section 4.7).
- 2 **Configure CAN.** 1 Mbit/s, standard 11-bit identifiers.
- 3 **Start the heartbeat.** Send LCI_HostHeartbeat (0x5A5) every 20 ms. Without it the chassis will not move under host control (section 5.2.2).
- 4 **Hand over control.** With the transmitter on and enabled, set switch C down. The chassis now follows your commands.
- 5 **Send commands.** Start with low speeds. The ROS software (section 5.4) is the quickest way to get going: it sends the heartbeat for you.

KEEP A WAY TO STOP

While your software is under development, keep the transmitter in hand. Flipping switch C up takes control back immediately; switch A stops the vehicle.



FUNCTIONS IN DETAIL

4.1 Power-on self-test

At switch-on the chassis controller checks the battery management system, the four motor controllers, both steering units, the brake and the parking brake before it allows the vehicle to move. Faults are shown on the interface (section 5.4) and the vehicle stays parked.

4.2 Controls on the chassis

Table 4-1 · Side control panel (Figure 2-2)

No.	Item	Function
1	Battery display	Battery voltage, state of charge and current. Lit while the chassis is on.
2	Start button	Switches the chassis on and off.
3	Emergency stop	Stops the vehicle (section 4.5). Turn clockwise to release.
4	Charge indicator	Shows charging progress.
5	Service / CAN port	Weatherproof 5-pin connector with the CAN bus (section 4.7).
6	Payload power port	24 V output for your equipment (section 4.7).
7	Payload power switch	Switches the 24 V payload output on and off.
8	Charging inlet	AC charging connection, with a weatherproof cover.

4.3 Remote control

The transmitter controls the vehicle only when all four conditions are met: the receiver has a valid signal, the transmitter is online, switch C is set to remote control, and switch F is set to enable.

Table 4-2 · Driving with the transmitter

Control	Behaviour
Right stick up	Drive. Speed rises with stick travel up to the limit set with knob VRC. The vehicle only drives with a gear selected, a speed limit above zero, no brake demand, no emergency stop and no motor or battery fault.
Right stick down	Brake, proportional to stick travel, up to full brake pressure (6 MPa).
Left stick	Steering, proportional to stick travel. Steering speed falls as road speed rises: about 520 °/s at standstill down to about 130 °/s at top speed.
Both sticks	A dead band of $\pm 5\%$ around centre is ignored.
Knob VRC	Sets the top speed for remote control. Forward 0–30 km/h, reverse limited to 10 km/h in current firmware.

4.4 Control modes

The chassis always knows who is in control, and it, not the host computer, decides which source it accepts.

Table 4-3 · Control modes

Mode	How to enter	Accepts	Purpose
Remote control	Transmitter online, switch F enable, switch C up	Transmitter	Manual driving, moving the vehicle, supervising tests.
Automatic	Transmitter online, switch F enable, switch C down, host heartbeat present	Host computer	Your autonomy software drives the vehicle over LCI.
Maintenance	Planned	Neither	Planned: motion locked out; used to clear faults and recover after an emergency stop without switching off.
No control	No valid source	Neither	If moving, the vehicle brakes for 5 s to a stop and applies the parking brake; if stationary, it parks immediately.

REMOTE CONTROL ALWAYS WINS

The transmitter link is independent of the host interface. Setting switch C up, or switch A down, takes effect regardless of what the host computer is sending.

4.5 Safety functions

4.5.1 Emergency stop

Any of the following triggers an emergency stop:

- the emergency stop button on the side panel,
- the front or rear bumper strip,
- switch A on the transmitter,
- an emergency stop request from the host computer (LCI_HostHeartbeat, EStopRequest),
- an external hard-wired stop loop on the payload connector (planned).

On an emergency stop the vehicle brakes at full pressure for 5 seconds and then applies the parking brake. The chassis reports which source triggered the stop (LCI_ChassisState, EStopSource).

Table 4-4 · Recovering from an emergency stop

Firmware	Recovery
Current	Remove the cause, release the button, then switch the chassis off and on again.
Planned (LCI)	Remove the cause, release the button, switch to maintenance mode, then back. No power cycle.

4.5.2 Bumpers

After a front-bumper contact the vehicle brakes for 5 seconds and can then only reverse. After a rear-bumper contact it brakes for 5 seconds and can then only drive forward.

4.5.3 Host heartbeat

In automatic mode the chassis expects LCI_HostHeartbeat every 20 ms. If it is missing for 200 ms, or its counter stops changing, the chassis brings the vehicle to a controlled stop, applies the parking brake and reports a heartbeat timeout. Available with the LCI firmware.

4.5.4 Live limits

The chassis limits speed and steering for slope, roll, temperature, battery state and faults. Commands above a limit are executed at the limit, not rejected, and the chassis reports the current limits and whether a command was clipped (LCI_LimitsFb). Available with the LCI firmware.

4.5.5 Parking brake

The electric parking brake applies automatically when the vehicle stops without a control source and after an emergency stop. It holds a fully loaded vehicle on a 20° slope for at least 5 minutes.

4.6 Lights

Table 4-5 · Lights

Light	Behaviour
Headlights	Switch D on the transmitter, or LCI_LampCmd from the host.
Turn indicators	Come on automatically when the vehicle steers left or right; can also be set by the host (including hazard lights).
Brake lights	Automatic with brake demand.
Reverse lights	Automatic in reverse gear.
Stealth mode	Switches all lights off, by remote control or by command.
Status light	Planned: yellow = maintenance, green = remote control, blue = automatic, as on Luchs A.

4.7 Payload interfaces

Table 4-6 · Connectors on the side panel

Port	Connector	Purpose	Pin assignment
Service / CAN port	Weatherproof circular, 5-pin (WS20 series)	CAN bus	Pin 1: reserved Pin 2: reserved Pin 3: CAN H Pin 4: CAN L Pin 5: not used Shielded twisted pair. Not terminated at the chassis end.
Payload power port	Weatherproof circular, 2-pin (WS20 series)	24 VDC for payload	Pin 1: 24 V + Pin 2: 24 V – Rating: 12 V, 25 A nominal (300 W), 30 A maximum.

Deck

The flat deck carries sensor masts, compute and cargo up to the payload limits in section 2.3.

COMMUNICATION INTERFACES AND SOFTWARE

5.1 Overview

Table 5-1 · Interfaces for your software

Interface	Use	Documentation
CAN, LCI v1.0, 1 Mbit/s	Main interface: motion, steering, heartbeat, live limits, all feedback	Section 5.2 · Luchs_B_CAN_Protocol_LCI_v1.0_draft.xlsx, Luchs_B_CAN_LCI_v1.0_draft.dbc
CAN, VCU interface V2.6	Units shipped before the LCI firmware	Section 5.3 · Luchs_B_CAN_Protocol_VCU_V2.6.xlsx, Luchs_B_CAN_VCU_V2.6.dbc
ROS 2 / ROS 1	Ready-made driver on either interface	Section 5.4 · luchs_ros2_v1.1.0.zip, luchs_ros1_v1.1.0.zip

5.2 CAN protocol (LCI)

LCI is the CAN interface of the Luchs family; its core frames use the same identifiers and layouts as the Luchs A protocol.

PRELIMINARY

LCI v1.0 is a draft and the Luchs B firmware that implements it is in development. Message IDs and layouts may still change.

Standard	CAN 2.0B, standard 11-bit identifiers, 8-byte payload
Bit rate	1 Mbit/s, on a dedicated customer CAN channel
Byte order	Big-endian (Motorola): the high byte of a multi-byte signal is sent first. Signed values in two's complement.
Physical value	Raw value × factor + offset
Command level	Motion level only: target speed with a yaw rate, or with front and rear road-wheel angles, plus an optional acceleration limit. The chassis closes the speed, steering and brake loops. There are no throttle, brake-pressure or steering-motor commands.
Timing	Host commands every 20 ms (any period up to 50 ms is accepted). Feedback every 20 ms; IMU every 10 ms.
Who is in charge	The host states intent; the chassis is the final authority on modes, limits and stopping.

5.2.1 Reference frame and motion models

All motion values refer to the **rear-axle centre** on the ground, with X forward, Y left and Z up (ROS REP 103). Yaw is positive anticlockwise seen from above.

Table 5–2 · Motion models on Luchs B (KinematicModel)

Value	Model	Steering from yaw rate
0x02	Front Ackermann	$\delta_f = \text{atan}(L \cdot \omega_z / v_x)$, rear axle straight
0x04	Dual Ackermann	$\delta_f = -\delta_r = \text{atan}(L \cdot \omega_z / (2 v_x))$. The rear-axle centre moves sideways: $v_y = -\omega_z \cdot L / 2$, reported in ActualVy.
0x01	Skid-steer	Not offered on Luchs B

L is the wheelbase, 1.430 m. Set **SteerSource** in LCI_SteerCmd to 1 to command road-wheel angles directly instead of a yaw rate; this also works at standstill.

5.2.2 Heartbeat and timeouts

- Send LCI_HostHeartbeat (0x5A5) every 20 ms, increment AliveCounter by 1 in every frame (255 wraps to 0), and set Checksum to the low byte of the sum of bytes 0–6, XOR 0xFF.
- If the heartbeat is missing for 200 ms, or AliveCounter stops changing, the chassis stops under control, parks and sets EStopSource bit 7.
- EStopRequest** is always executed as an emergency stop. Use **ParkRequest** when you want the chassis to stop under control.

5.2.3 Message definitions

Table 5–3 · LCI messages on Luchs B

CAN ID	Message	Direction	Cycle	Use on Luchs B
0x5A0	LCI_MotionCmd	Host → Chassis	20 ms nominal (any period $\leq$ 50 ms accepted)	Luchs B
0x5A1	LCI_SteerCmd	Host → Chassis	20 ms nominal ($\leq$ 50 ms)	Luchs B
0x5A3	LCI_LampCmd	Host → Chassis	100 ms	Luchs B
0x5A4	LCI_HostTime	Host → Chassis	1 s	Luchs B
0x5A5	LCI_HostHeartbeat	Host → Chassis	20 ms (timeout 200 ms)	Luchs B
0x5B0	LCI_LimitsFb	Chassis → Host	20 ms	Luchs B
0x5B1	LCI_ImuAccelFb	Chassis → Host	10 ms	Optional (with chassis IMU)
0x5B2	LCI_AttitudeFb	Chassis → Host	20 ms	Optional (with chassis IMU)
0x5B5	LCI_MotionFb	Chassis → Host	20 ms	Luchs B
0x5B6	LCI_SteerFb	Chassis → Host	20 ms	Luchs B
0x5B8	LCI_ProtectionFb	Chassis → Host	50 ms	Luchs B
0x5B9	LCI_WheelSpeedFb	Chassis → Host	20 ms	Luchs B
0x5BA	LCI_MotorFault	Chassis → Host	100 ms	Luchs B
0x5BB	LCI_BatteryFb	Chassis → Host	100 ms	Luchs B
0x5BD	LCI_ChassisState	Chassis → Host	20 ms	Luchs B

CANID	Message	Direction	Cycle	Use on Luchs B
0x5BE	LCI_ImuRateFb	Chassis → Host	10 ms	Optional (with chassis IMU)
0x5C0	LCI_B_BrakeSteerFb	Chassis → Host	50 ms	Luchs B
0x5C1	LCI_B_MotorTempFb	Chassis → Host	500 ms	Luchs B
0x5BF	LCI_Version	Chassis → Host	1s	Luchs B

All frames are available with the LCI firmware. The four-wheel-steer frames 0x5A2 and 0x5B7 are not used on Luchs B.

Table 5-4 · LCI message definitions

Position	Signal	Type	Scaling	Values and notes
0X5A0 LCI_MOTIONCMD · HOST → CHASSIS · 20 MS NOMINAL (ANY PERIOD ≤ 50 MS ACCEPTED)				
Body-frame velocity command at the reference point (rear-axle centre, X forward, Y left). The chassis closes the speed and yaw loops and clamps to its live limits (0x5B0).				
bytes 0-1	TargetVx	int16	0.001 m/s	Longitudinal speed, +forward
bytes 2-3	TargetVy	int16	0.001 m/s	Lateral speed, +left. 4WS only; others send 0
bytes 4-5	TargetWz	int16	0.001 rad/s	Yaw rate, +counter-clockwise
byte 6	KinematicModel	uint8	1	1=Skid-steer 2=Ackermann (front) 3=4WS 4=Dual Ackermann
byte 7	ControlMode	uint8	1	0x00 = autonomous (only value)
0X5A1 LCI_STEERCMD · HOST → CHASSIS · 20 MS NOMINAL (≤ 50 MS)				
Gear and axle steering angles for Ackermann-type chassis.				
byte 0	Gear	uint8	1	0=disabled 1=P 2=R 3=N 4=D
bytes 1-2	FrontSteerAngle	int16	0.001 rad	Road-wheel angle, front axle, +left
bytes 3-4	RearSteerAngle	int16	0.001 rad	Road-wheel angle, rear axle, +left (dual Ackermann / AWS)
byte 5	SteerSource	uint8	1	r3. 0 = steer from TargetWz in 0x5A0 (yaw-rate mode) · 1 = steer to the angles in this frame (angle mode, also at standstill)
bytes 6-7	AccelLimit	uint16	0.001 m/s ²	r3. Max acceleration to reach TargetVx; 0 = chassis default ramp
0X5A3 LCI_LAMPCMD · HOST → CHASSIS · 100 MS				
Lamps and horn. Brake and reverse lamps stay automatic on chassis that drive them themselves.				
byte 0	TurnIndicator	uint8	1	0=off 1=left 2=right 3=hazard (3 new in r3)
byte 1	ReverseLamp	uint8	1	0=normal 1=reversing (ignored where automatic)
byte 2	FaultLamp	uint8	1	0=normal 1=host system fault
byte 3	Headlight	uint8	1	r3. 0=off 1=low 2=high (moved here from 0x5A5 bit 10)
byte 4	Horn	uint8	1	r3. 1=sound
byte 5	Beacon	uint8	1	r3. Warning beacon, where fitted
0X5A4 LCI_HOSTTIME · HOST → CHASSIS · 1 S				

Position	Signal	Type	Scaling	Values and notes
Host wall-clock time so chassis logs and fault records can be matched to host logs.				
bytes 0-7	HostTimeMs	uint64	1ms	Milliseconds since 1970-01-01 UTC
0X5A5 LCI_HOSTHEARTBEAT · HOST → CHASSIS · 20 MS (TIMEOUT 200 MS)				
Liveness and safety requests from the host. Missing or frozen for 200 ms → controlled stop, park, report HeartbeatTimeout. EStopRequest is always executed as an emergency stop and never downgraded; ParkRequest is the controlled stop.				
byte 0	AliveCounter	uint8	1	Increments by 1 every frame, wraps 255→0
bit 8	EStopRequest	bit	1	bit 8. 1 = emergency stop (latched until maintenance reset)
bit 9	ParkRequest	bit	1	bit 9. 1 = stop and hold
bit 11	PayloadPower1_On	bit	1	bit 11. Switched payload output 1
bit 12	PayloadPower2_On	bit	1	bit 12. Switched payload output 2
bytes 2-3	SpeedLimit	uint16	0.01 m/s	Host-side speed cap; 0 = use chassis default
byte 7	Checksum	uint8	1	Low byte of (sum of bytes 0-6), XOR 0xFF
0X5B0 LCI_LIMITSFB · CHASSIS → HOST · 20 MS				
Live limits the chassis will enforce right now (slope, roll, temperature, battery, faults, host cap). Commands beyond them are executed at the limit and flagged.				
bytes 0-1	MaxVxFwd	uint16	0.01 m/s	Allowed forward speed
bytes 2-3	MaxVxRev	uint16	0.01 m/s	Allowed reverse speed
bytes 4-5	MaxSteerAngle	uint16	0.001 rad	Allowed road-wheel angle at the current speed
byte 6	LimitReason	uint8	1	bit field: 0 slope/roll · 1 temperature · 2 battery · 3 fault derate · 4 host SpeedLimit · 5 speed-dependent steer limit · 6 maintenance/RC · 7 other
bit 56	SpeedClipped	bit	1	bit 56. Last TargetVx exceeded the limit and was clamped
bit 57	SteerClipped	bit	1	bit 57. Last steering request was clamped
bit 58	AccelClipped	bit	1	bit 58. Last AccelLimit was clamped
0X5B1 LCI_IMUACCELFB · CHASSIS → HOST · 10 MS				
Chassis IMU accelerations in the vehicle frame (X forward, Y left, Z up), gravity included. Paired with 0x5BE by ImuCounter.				
bytes 0-1	AccelX	int16	0.002 m/s^2	
bytes 2-3	AccelY	int16	0.002 m/s^2	
bytes 4-5	AccelZ	int16	0.002 m/s^2	≈ +9.81 at rest on level ground
byte 6	ImuCounter	uint8	1	Same value as the matching 0x5BE frame
0X5B2 LCI_ATTITUDEFB · CHASSIS → HOST · 20 MS				
Roll and pitch estimated by the chassis; the same values it uses for slope and roll-over limits.				
bytes 0-1	Roll	int16	0.0001 rad	+right side down
bytes 2-3	Pitch	int16	0.0001 rad	+nose down
0X5B5 LCI_MOTIONFB · CHASSIS → HOST · 20 MS				

Position	Signal	Type	Scaling	Values and notes
Measured body-frame motion at the reference point (rear-axle centre). In dual Ackermann the rear-axle centre moves sideways: ActualV _y = -ActualW _z ·L/2.				
bytes 0-1	ActualV _x	int16	0.001 m/s	
bytes 2-3	ActualV _y	int16	0.001 m/s	4WS and dual Ackermann; 0 otherwise
bytes 4-5	ActualW _z	int16	0.001 rad/s	
byte 6	KinematicModel	uint8	1	As 0x5A0
byte 7	ControlMode	uint8	1	0x11 = remote control, 0x22 = autonomous (hexadecimal)
0X5B6 LCI_STEERFB · CHASSIS → HOST · 20 MS				
Measured gear and axle steering angles.				
byte 0	Gear	uint8	1	As 0x5A1
bytes 1-2	FrontSteerAngle	int16	0.001 rad	
bytes 3-4	RearSteerAngle	int16	0.001 rad	
0X5B8 LCI_PROTECTIONFB · CHASSIS → HOST · 50 MS				
Proximity and contact protection.				
byte 0	UltrasonicStop	uint8	1	0=none 1=left 2=right 3=front 4=rear
byte 1	BumperHit	uint8	1	0=none 1=front 2=rear
byte 2	EStopButton	uint8	1	0=none 1=left 2=right
byte 3	FaultCode	uint8	1	Chassis fault code (Appendix B)
byte 4	UltrasonicDist_Front	uint8	1 cm	1 cm per bit; 255 = no echo / not fitted
byte 5	UltrasonicDist_Rear	uint8	1 cm	
byte 6	UltrasonicDist_Left	uint8	1 cm	
byte 7	UltrasonicDist_Right	uint8	1 cm	
0X5B9 LCI_WHEELSPEEDFB · CHASSIS → HOST · 20 MS				
Drive-wheel angular speeds. Order: LF, RF, LR, RR. 0.01 rad/s covers both chassis (B ≈ 40 rad/s at 40 km/h).				
bytes 0-1	WheelSpeed_LF	int16	0.01 rad/s	
bytes 2-3	WheelSpeed_RF	int16	0.01 rad/s	
bytes 4-5	WheelSpeed_LR	int16	0.01 rad/s	
bytes 6-7	WheelSpeed_RR	int16	0.01 rad/s	
0X5BA LCI_MOTORFAULT · CHASSIS → HOST · 100 MS				
Drive and steering motor fault bit fields.				
bytes 0-3	MotorFaultHigh	uint32	1	Bit table to be published
bytes 4-7	MotorFaultLow	uint32	1	Bit table to be published
0X5BB LCI_BATTERYFB · CHASSIS → HOST · 100 MS				

Position	Signal	Type	Scaling	Values and notes
Battery state. Scalings fixed so no field overflows on a 48 V or 72 V pack.				
bytes 0-1	SOC	uint16	0.01%	0.01% per bit
bytes 2-3	BatteryTemp	int16	0.1 °C	0.1 °C per bit
bytes 4-5	ChargeProgress	uint16	1%	
bytes 6-7	BatteryVoltage	uint16	0.1 V	0.1 V per bit (A: confirmed by measurement)
0X5BD LCI_CHASSISSTATE · CHASSIS → HOST · 20 MS				
Who is in control, why it stopped, and whether it is alive.				
byte 0	OperatingMode	uint8	1	0=maintenance 1=remote control 2=automatic
byte 1	EStopSource	uint8	1	bit field: 0 onboard L · 1 onboard R · 2 RC · 3 host · 4 bumper F · 5 bumper R · 6 external loop · 7 heartbeat timeout
byte 2	FaultLevel	uint8	1	0=none 1=warning 2=derated 3=stop
bit 24	ROnline	bit	1	bit 24
bit 25	RCEnabled	bit	1	bit 25
bit 26	DriveMode4WD	bit	1	bit 26. 1=4WD 0=2WD (B)
bit 27	ParkBrakeEngaged	bit	1	bit 27. EPB (B) or motor hold (A) engaged
bits 28-29	RCLink	bit	1	bits 28-29. r3. 0=good 1=weak 2=lost 3=not fitted
bit 30	CmdClipped	bit	1	bit 30. r3. Any command clamped (details in 0x5B0)
bytes 4-5	ActiveSpeedLimit	uint16	0.01 m/s	Effective cap after all limits (= 0x5B0 MaxVxFwd)
byte 6	AliveCounter	uint8	1	
byte 7	Checksum	uint8	1	Low byte of (sum of bytes 0-6), XOR 0xFF
0X5BE LCI_IMURATEFB · CHASSIS → HOST · 10 MS				
Chassis IMU angular rates in the vehicle frame. 100 Hz, as asked for in the chassis-side July reply.				
bytes 0-1	RollRate	int16	0.001 rad/s	+right side down
bytes 2-3	PitchRate	int16	0.001 rad/s	+nose down
bytes 4-5	YawRate	int16	0.001 rad/s	+counter-clockwise
byte 6	ImuCounter	uint8	1	Increments every sample
byte 7	ImuStatus	uint8	1	0=ok 1=initialising 2=fault
0X5C0 LCI_B_BRAKESTEERFB · CHASSIS → HOST · 50 MS				
Brake, parking-brake and steering-actuator status for diagnostics.				
bytes 0-1	BrakePressure	uint16	0.01 MPa	Master-cylinder pressure from the brake booster
byte 2	EpbState	uint8	1	0=released 1=applying 2=applied 3=releasing 4=fault
byte 3	EpsFrontState	uint8	1	0=ok 1=warning 2=degraded 3=failed
byte 4	EpsRearState	uint8	1	0=ok 1=warning 2=degraded 3=failed

Position	Signal	Type	Scaling	Values and notes
byte 5	BrakeState	uint8	1	0=ok 1=warning 2=degraded 3=failed
0X5C1 LCI_B_MOTORTEMPFB · CHASSIS → HOST · 500 MS				
Hub-motor and motor-controller temperatures (offset -40 °C).				
byte 0	MotorTemp_LF	uint8	1°C, offset -40	
byte 1	MotorTemp_RF	uint8	1°C, offset -40	
byte 2	MotorTemp_LR	uint8	1°C, offset -40	
byte 3	MotorTemp_RR	uint8	1°C, offset -40	
byte 4	CtrlTemp_LF	uint8	1°C, offset -40	
byte 5	CtrlTemp_RF	uint8	1°C, offset -40	
byte 6	CtrlTemp_LR	uint8	1°C, offset -40	
byte 7	CtrlTemp_RR	uint8	1°C, offset -40	
0X5BF LCI_VERSION · CHASSIS → HOST · 1 S				
Lets one driver identify the chassis and protocol version at start-up.				
byte 0	LciMajor	uint8	1	
byte 1	LciMinor	uint8	1	
byte 2	ModelId	uint8	1	0x0A A-DF · 0x0B A-FA · 0x0C A-DA · 0x0D A-4WS · 0x1B B (4 hub motors, dual EPS, single/dual Ackermann) · 0x2C C
byte 3	FwMajor	uint8	1	
byte 4	FwMinor	uint8	1	
byte 5	FwPatch	uint8	1	

5.3 VCU interface V2.6

Luchs B units shipped before the LCI firmware use the upper-computer interface of the vehicle control unit. It works at pedal level: your software sets the accelerator and brake, the road-wheel angle and a speed cap, and the chassis reports its speed in whole km/h. It is suited to bring-up, teleoperation and data recording; for autonomy use LCI. The ROS software supports it with `protocol:=legacy_b` and closes the speed loop for you.

Table 5-5 · VCU interface V2.6

Message	Direction · cycle	Content
0x701 UpperCmd	Host → Chassis · 10 ms	Accelerator and brake (%), road-wheel angle ($\pm 42^\circ$), speed cap (km/h), steering rate, upper-computer control, headlight, emergency stop, park, gear (N / D / R), heartbeat
0x703 VCU2Upper	Chassis → Host · 10 ms	State of charge, battery temperature and voltage, fault level, gear, mode, speed (km/h), heartbeat

Byte order little-endian (Intel). Before the first drive, check the bit rate (500 kbit/s) and the sign of the steering angle on your vehicle; the ROS software sets both by parameter. Full definitions:

Luchs_B_CAN_Protocol_VCU_V2.6.xlsx and Luchs_B_CAN_VCU_V2.6.dbc.

5.4 ROS software

The ROS software for the Luchs family is supplied as `luchs_ros2` (version 1.1, ROS 2 Humble and Jazzy) and `luchs_ros1` (version 1.1, ROS 1 Noetic). Both are built on the same ROS-independent C++ SDK and drive Luchs B and Luchs A from one code base, with the same topics, frames and services. Publish a velocity on `cmd_vel`; the driver sends the LCI commands and the heartbeat every 20 ms, applies the live limits the chassis reports (keeping the path curvature), and stops when commands or feedback stop. If the driver process stops, the heartbeat stops and the chassis parks by itself within 200 ms.

Table 5–6 · Main topics and services of the driver

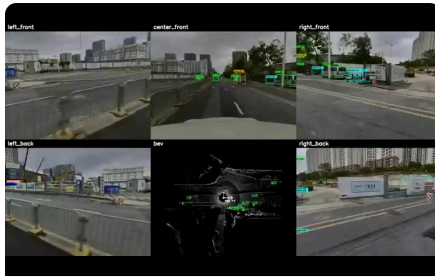
Direction	Topic / service	Type	Content
Subscribed	<code>cmd_vel</code>	<code>geometry_msgs/Twist</code>	Velocity: forward and yaw rate (dual or front Ackermann)
Published	<code>odom</code>	<code>nav_msgs/Odometry</code>	Odometry at <code>base_link</code> and the <code>odom</code> → <code>base_footprint</code> transform
Published	<code>imu/data_raw</code>	<code>sensor_msgs/Imu</code>	Chassis IMU (LCI 0×5BE / 0×5B1)
Published	<code>~/state</code> · <code>~/safety</code>	<code>luchs_msgs/...</code>	Control mode, e-stop source, live limits, EPB / EPS / brake, temperatures, model and firmware
Published	<code>joint_states</code> · <code>battery_state</code> · <code>/diagnostics</code>	<code>sensor_msgs/...</code> · <code>diagnostic_msgs/...</code>	Steering and wheel joints, battery, link and fault status
Service	<code>~/estop</code> · <code>~/park</code>	<code>std_srvs/SetBool</code>	Emergency-stop and park requests in the heartbeat
Service	<code>~/set_speed_limit</code> · <code>~/set_payload_power</code>	<code>luchs_msgs/...</code>	Host speed cap; switched payload outputs

Quick start (ROS 2): bring the CAN interface up at 1 Mbit/s (`setup_can.sh can0`), set switch C down, then `ros2 launch lushs_bringup lushs_b.launch.py`; units with the VCU interface: add `protocol:=legacy_b`. **ROS 1:** `roslaunch lushs_bringup lushs_b.launch`. Without a vehicle, add `dry_run:=true` to run against the built-in simulator, which follows LCI including the heartbeat timeout. The README and the docs folder in each package cover installation, wiring, parameters and troubleshooting.

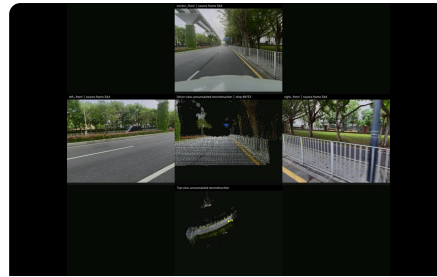
The software has been validated against the built-in simulator.

5.5 Autonomy by TAR (option)

The Luchs family is built for autonomy. TerraAgent Robotics (TAR) supplies it as an option: sensor kit, compute, positioning, perception, planning, vehicle control and the control station, integrated on the Luchs B and delivered as one system. TAR software drives the chassis through the interfaces and the ROS driver described in this chapter, so sections 5.1 to 5.4 apply to a vehicle with the TAR kit as well. Figure 5–1 shows building blocks that already run on real vehicles and real data.



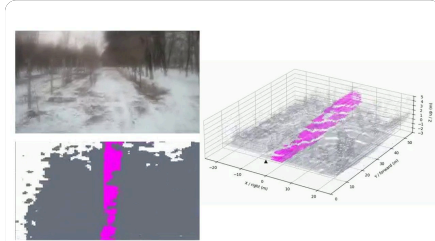
Camera-first perception
REAL VEHICLE



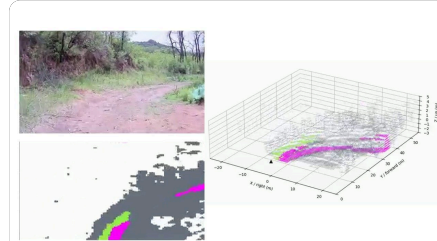
Dense 3D reconstruction
REAL VEHICLE



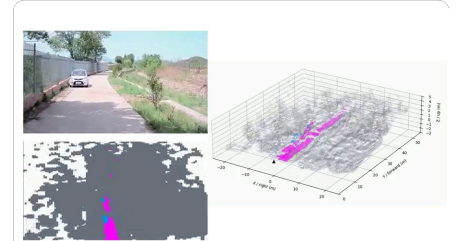
Reconstruction against low sun
REAL VEHICLE



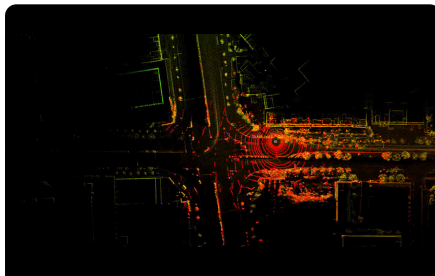
Drivable ground in snow
RECORDED OFF-ROAD DATA,
PROCESSED OFFLINE



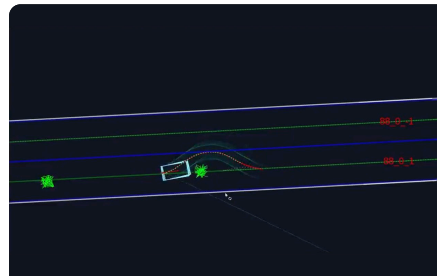
Dirt trail and verge
RECORDED OFF-ROAD DATA,
PROCESSED OFFLINE



Oncoming vehicle
RECORDED OFF-ROAD DATA,
PROCESSED OFFLINE



LiDAR map localisation
REAL VEHICLE, REAL TIME



Planning around an obstacle
PLANNING STACK, REPLAY



Junctions, U-turns and parking
PLANNING STACK, REPLAY

Figure 5-1 · TAR building blocks. Top: camera perception and 3D reconstruction on real vehicles. Middle: off-road drivable ground, our model run offline on recorded off-road driving data. Bottom: LiDAR map localisation on a real vehicle, and replays of the planning stack.

- **Positioning** degrades safely: RTK, then visual-inertial coasting at reduced speed, then a timed safe stop; LiDAR map localisation is the second source when satellites drop out.
- **Perception** is camera-first, with LiDAR and radar added per application: 3D detection, a drivable-ground map and a live elevation map for holes and steps.
- **Planning** runs in three layers (route, behaviour, trajectory) and hands its commands to the chassis through the ROS driver.

The functions, the sensor kit and the delivery for your site are agreed with TAR. Contact details are in section 6.3.



MAINTENANCE AND SUPPORT

6.1 Routine care

- Check tyre condition before each use.
- Keep the battery between 50 % and 70 % for storage longer than a month, and charge it at least every 3 months.
- Clean mud and sand from the wheels, suspension and bumper strips after off-road use. Do not use a high-pressure washer on connectors or controllers.
- Battery swap takes under 3 minutes with the correct tools.

6.2 Warranty

Chassis: 12 months or 2000 km off-road / 4000 km paved, whichever comes first. Repairs: 90 days on the same issue. Excludes normal wear, consumables and damage from misuse, accident or unauthorised modification.

6.3 Support

CHASSIS

RJ Tech Limited

Hardware, drivetrain, firmware, spare parts and warranty.

rjtech-offroad.com

AUTONOMY

TerraAgent Robotics

Sensors, compute, autonomy software and integration.

Contact: Brandon Zhe Feng

When you contact support, please have the serial number, the firmware version (LCI_Version) and a description of the problem ready, including any fault reported by the chassis.

DIMENSIONS



All dimensions in millimetres.

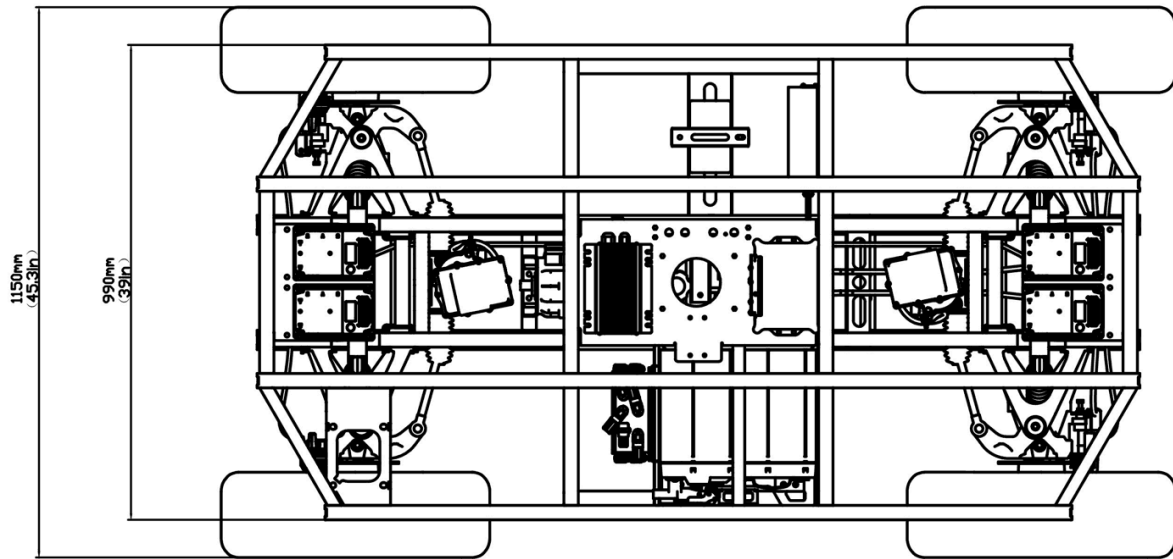


Figure A-1 · Top view

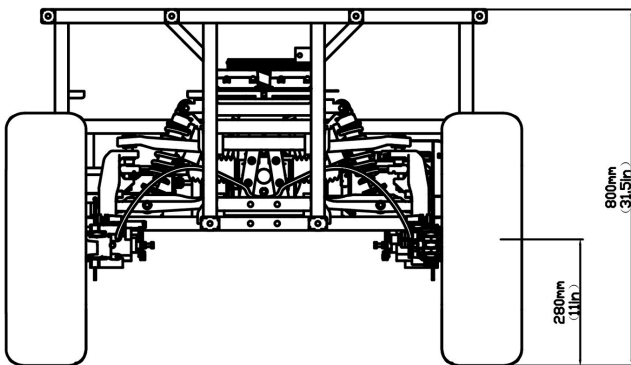


Figure A-2 · Front view

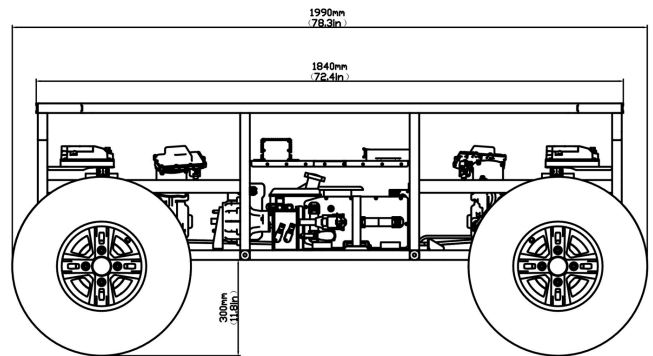


Figure A-3 · Side view



FAULT CODES

The chassis reports a fault level in LCI_ChassisState (0 none, 1 warning, 2 derated, 3 stop) and detailed fault bits in LCI_MotorFault.



TERMS AND ABBREVIATIONS

2WD / 4WD	Two- or four-wheel drive.
Ackermann steering	Car-style steering in which the inner wheel turns more sharply than the outer one, so both follow the same arc.
BMS	Battery Management System.
CAN	Controller Area Network: a robust two-wire bus used in vehicles and machines.
DBC	A standard file format describing CAN messages and signals.
Dual Ackermann	Both axles steer, in opposite directions, for a tighter turn.
EPB	Electric Parking Brake.
EPS	Electric Power Steering.
Heartbeat	A message the host sends at a fixed rate so the chassis knows it is alive.
IMU	Inertial Measurement Unit: measures rotation rates and accelerations.
LCI	Luchs Chassis Interface.
Odometry	An estimate of the vehicle's motion from wheel speeds and steering angles.
REP103	The ROS convention for units and coordinate frames.
ROS 2	Robot Operating System 2.
Yawrate	How fast the vehicle turns about its vertical axis.